

Опр. Арифметическое н-во $\mathbb{R}^n$

$$X(x_1, \dots, x_n) \quad x_1, \dots, x_n \in \mathbb{R}$$

$$\bar{x} + \bar{y} (x_1 + y_1, \dots, x_n + y_n)$$

$$\alpha \in \mathbb{R} \quad \alpha \bar{x} (\alpha x_1, \dots, \alpha x_n)$$

Умб: $\mathbb{R}^n$ - линейное н-во

Лек-во: **непосредственная проверка аксиом**

Опр. линейное век. н-во X нз Евклидова, если на нем задан ф-л $(\dots): [X \times X] \rightarrow \mathbb{R}$

скалярн. произв.

$$1) \forall \bar{x} \in X \quad (\bar{x}; \bar{x}) \geq 0 \quad (\bar{x}; \bar{x}) = 0 \Leftrightarrow \bar{x} = 0$$

$$2) \forall \bar{x}, \bar{y} \in X \quad (\bar{x}; \bar{y}) = (\bar{y}; \bar{x})$$

$$3) \forall \alpha, \beta \in \mathbb{R} \quad \forall \bar{x}, \bar{y}, \bar{z} \in X \quad \alpha \bar{x} + \beta \bar{y}; \bar{z} = \alpha (\bar{x}; \bar{z}) + \beta (\bar{y}; \bar{z})$$

Умб: $\mathbb{R}^n$ нз Евклидова н-ва, если

$$(\bar{x}; \bar{y}) = x_1 y_1 + \dots + x_n y_n$$

Умб $\forall \bar{x}, \bar{y} \in X \quad ((\bar{x}; \bar{y}))^2 \leq (\bar{x}; \bar{x})(\bar{y}; \bar{y})$ - н-во Коши-Буняковского - Евклидова

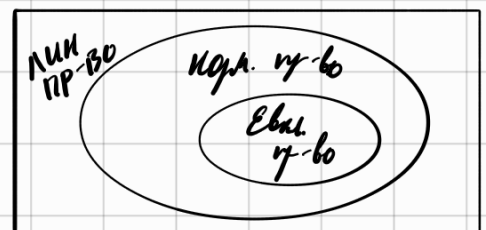
$$\Delta \quad 0 \leq (\alpha \bar{x} + \bar{y}; \alpha \bar{x} + \bar{y}) = \alpha (\bar{x}; \alpha \bar{x} + \bar{y}) + (\bar{y}; \alpha \bar{x} + \bar{y}) = \alpha^2 (\bar{x}; \bar{x}) + 2\alpha (\bar{x}; \bar{y}) + (\bar{y}; \bar{y})$$

$$D \quad ((\bar{x}; \bar{y}))^2 - (\bar{x}; \bar{x})(\bar{y}; \bar{y}) \leq 0$$

$$((\bar{x}; \bar{y}))^2 \leq (\bar{x}; \bar{x})(\bar{y}; \bar{y})$$

$$\text{Пусть } \bar{x} = 0 \quad 0 \leq 0$$

$$(x_1 y_1 + \dots + x_n y_n)^2 \leq (x_1^2 + \dots + x_n^2)(y_1^2 + \dots + y_n^2) \quad \Delta$$



Опр X лин. н-во

$\|\cdot\| : X \rightarrow \mathbb{R}$ - норм-е ф-ция

$$1) \forall \bar{x} \in X \quad \|\bar{x}\| \geq 0$$

$$2) \forall \alpha \in \mathbb{R} \quad \forall \bar{x} \in X \quad \|\alpha \bar{x}\| = |\alpha| \cdot \|\bar{x}\|$$

$$3) \forall \bar{x}, \bar{y} \in X \quad \|\bar{x} + \bar{y}\| \leq \|\bar{x}\| + \|\bar{y}\|$$

Опр $(X, \|\cdot\|)$ - нормированное н-во

Умб X Евклидова $\Rightarrow X$ нормированное

$$\Delta \quad f(\bar{x}) = \sqrt{(\bar{x}; \bar{x})}$$

1) +

$$2) f(\alpha \bar{x}) = (\alpha \bar{x}; \alpha \bar{x})' = \alpha^2 (\bar{x}; \bar{x})' = |\alpha| \cdot f(\bar{x}) +$$

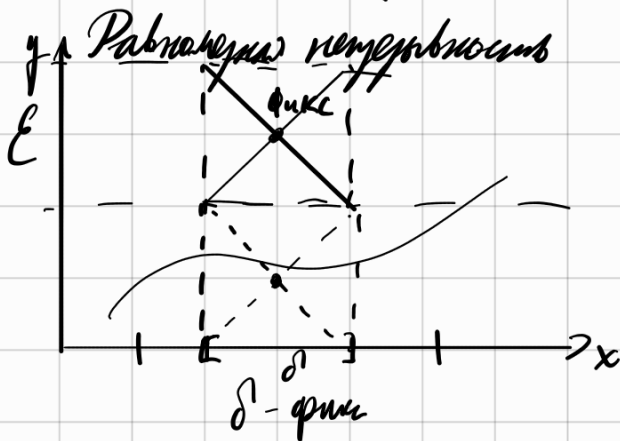
$$3) f(\bar{x} + \bar{y}) = (\bar{x} + \bar{y}; \bar{x} + \bar{y})' = (\bar{x}; \bar{x}) + 2(\bar{x}; \bar{y}) + (\bar{y}; \bar{y}) \leq \\ \leq (\bar{x}; \bar{x}) + 2\sqrt{(\bar{x}; \bar{x})(\bar{y}; \bar{y})} + (\bar{y}; \bar{y}) = (f(\bar{x}) + f(\bar{y}))' = f(\bar{x}) + f(\bar{y}) \quad \Delta$$

О-е: $\mathbb{R}^n$ - норм. н-во с нормой $\|\bar{x}\| = \sqrt{x_1^2 + \dots + x_n^2}$

Равенства: $|\bar{x}| = \|\bar{x}\| = \sqrt{x_1^2 + \dots + x_n^2}$

Опр. $\bar{x}_0$ - какая то точка $\in \mathbb{R}^n$ $\delta > 0$

$$U_\delta(\bar{x}_0) = \{\bar{x} \in \mathbb{R}^n : |\bar{x} - \bar{x}_0| < \delta\}$$



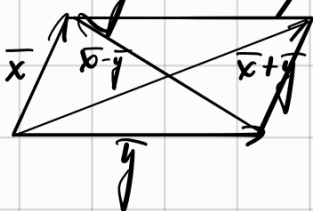
опр. н-во н-во с функ. центром, он равен
глубина (1/2 - < <) и он никогда не будет
пересекаясь с p-ей макс.

ТОЛЬКО ТАК



$$\|\bar{x} + \bar{y}\|^2 = (\bar{x} + \bar{y}; \bar{x} + \bar{y}) = \|\bar{x}\|^2 + 2(\bar{x}; \bar{y}) + \|\bar{y}\|^2 \\ \|\bar{x} - \bar{y}\|^2 = (\bar{x} - \bar{y}; \bar{x} - \bar{y}) = \|\bar{x}\|^2 - 2(\bar{x}; \bar{y}) + \|\bar{y}\|^2 \\ \|\bar{x} + \bar{y}\|^2 + \|\bar{x} - \bar{y}\|^2 = 2(\|\bar{x}\|^2 + \|\bar{y}\|^2)$$

Получено равенство.



$$\mathbb{R}^n \\ \|\bar{x}\| = \max_{k \in \{1, \dots, n\}} |x_k|$$

1) +

$$2) + \|\alpha \bar{x}\| = |\alpha| \|\bar{x}\|$$

$$3) + \|\bar{x} + \bar{y}\| \leq \|\bar{x}\| + \|\bar{y}\|$$

$$\max_k |x_k + y_k| \leq \max_k (|x_k| + |y_k|) \leq \max_k |x_k| + \max_k |y_k|$$

Опр. $f: U_\delta(\bar{x}_0) \rightarrow \mathbb{R}^n$

$$\forall \bar{x} \in U_\delta(\bar{x}_0) \exists ! \bar{y} \in \mathbb{R}^n : \bar{y} = f(\bar{x})$$

Задача вектор-ф-я

Def. $f: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$

$\lim_{x \rightarrow x_0} f(x) = \bar{a} \in \mathbb{R}^n$

по Коши

$\forall \varepsilon > 0 \exists \delta \in (0; \delta_0]: \forall x \in \dot{U}_{\delta}(x_0) \rightarrow f(x) \in U_{\varepsilon}(\bar{a}) \Leftrightarrow |f(x) - \bar{a}| < \varepsilon$

Def. $\lim_{k \rightarrow \infty} \bar{a}_k = \bar{a}$

$\forall \varepsilon > 0 \exists N: \forall k \geq N \rightarrow |\bar{a}_k - \bar{a}| < \varepsilon$

Def. $\lim_{x \rightarrow x_0} f(x) = \bar{a}$

$\forall \{x_k\}_{k=1}^{\infty} \subset \dot{U}_{\delta_0}(x_0): \lim_{k \rightarrow \infty} x_k = x_0 \rightarrow \lim_{k \rightarrow \infty} f(x_k) = \bar{a}$

Yml: $f: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n \quad \bar{a} \in \mathbb{R}^n$

Def. унитар экв.-ва

1) $\lim_{x \rightarrow x_0} f(x) = \bar{a}$

2) $\lim_{x \rightarrow x_0} f(f(x) - \bar{a}) = 0$

3) $\forall k \in \{1, \dots, n\} \rightarrow \lim_{x \rightarrow x_0} f_k(x) = a_k$

1) $\Leftrightarrow$ 2) по Коши

2) $\Leftrightarrow$ 3) $\sqrt{(f_1(x) - a_1)^2 + \dots + (f_n(x) - a_n)^2} < \varepsilon$

$|f_k(x) - a_k| \quad \forall k \in \{1, \dots, n\}$

$\forall \varepsilon > 0 \exists \delta \in (0; \delta_0]: \forall x \in \dot{U}_{\delta}(x_0) \rightarrow |f_k(x) - a_k| < \varepsilon$

3) $\Rightarrow$ 2) $\forall \varepsilon > 0 \exists \delta_k \in (0; \delta_0]: \forall x \in \dot{U}_{\delta_k}(x_0) \rightarrow |f_k(x) - a_k| < \frac{\varepsilon}{\sqrt{n}}$

$\forall \varepsilon > 0 \delta_{\min} \{ \delta_k \} \quad \forall x \in \dot{U}_{\delta}(x_0) \rightarrow \sum (f_k(x) - a_k)^2 < \varepsilon^2$
 $|f(x) - \bar{a}| < \varepsilon$ $\blacktriangle$

Yml $f: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$

$\lim_{x \rightarrow x_0} f(x) = \bar{a} \Rightarrow \lim_{x \rightarrow x_0} |f(x)| = |\bar{a}|$

$\blacktriangleright \|f(x) - \bar{a}\| \leq |f(x) - \bar{a}|$

$\forall \varepsilon > 0 \exists \delta \in (0; \delta_0]: \forall x \in \dot{U}_{\delta}(x_0) \rightarrow |f(x) - \bar{a}| < \varepsilon \quad \blacktriangle$

Yml $f: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$

$\varphi: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$

$\left. \begin{array}{l} \lim_{x \rightarrow x_0} f(x) = \bar{a} \\ \lim_{x \rightarrow x_0} \varphi(x) = A \end{array} \right\} \Rightarrow \lim_{x \rightarrow x_0} \varphi(x) f(x) = A \cdot \bar{a}$

$$\triangleright \varphi(x) \bar{f}(x) \left(\underbrace{\varphi(x)f_1(x)}_{\substack{\xrightarrow{x \rightarrow x_0} \\ Aa_1}}, \dots, \underbrace{\varphi(x)f_n(x)}_{\substack{\xrightarrow{x \rightarrow x_0} \\ Aa_n}} \right)$$

$$\lim_{x \rightarrow x_0} \varphi(x) \bar{f}(x) = A\bar{a} \quad \blacktriangle$$

$$\text{Ymb} \quad \bar{f}: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$$\bar{g}: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$$\left. \begin{array}{l} \lim_{x \rightarrow x_0} \bar{f}(x) = \bar{a} \\ \lim_{x \rightarrow x_0} \bar{g}(x) = \bar{b} \end{array} \right\} \Rightarrow \begin{array}{l} \lim_{x \rightarrow x_0} (\bar{f}(x) + \bar{g}(x)) = \bar{a} + \bar{b} \\ \lim_{x \rightarrow x_0} (\bar{f}(x); \bar{g}(x)) = (\bar{a}; \bar{b}) \end{array}$$

$$\triangleright (\bar{f}(x); \bar{g}(x)) = \sum_{k=1}^n \underbrace{f_k(x)}_{\substack{\xrightarrow{x \rightarrow x_0} \\ Aa_k}} g(x) \rightarrow \sum_{k=1}^n a_k b_k = (\bar{a}; \bar{b}) \quad \blacktriangle$$

Def. $\forall \bar{a}; \bar{b} \in \mathbb{R}^3$ означені

$$[\bar{a}; \bar{b}] (a_2 b_3 - a_3 b_2, a_3 b_1 - a_1 b_3, a_1 b_2 - a_2 b_1)$$

$$\begin{vmatrix} a_2 & a_3 \\ b_2 & b_3 \end{vmatrix} - \begin{vmatrix} a_1 & a_3 \\ b_1 & b_3 \end{vmatrix} \quad \begin{vmatrix} a_1 & a_2 \\ b_1 & b_2 \end{vmatrix}$$

$$\text{Ymb} \quad \bar{f}: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^3$$

$$\bar{g}: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^3$$

$$\left. \begin{array}{l} \lim_{x \rightarrow x_0} \bar{f}(x) = \bar{a} \\ \lim_{x \rightarrow x_0} \bar{g}(x) = \bar{b} \end{array} \right\} \Rightarrow \lim_{x \rightarrow x_0} [\bar{f}(x); \bar{g}(x)] = [\bar{a}; \bar{b}]$$

$$\text{Def} \quad \bar{f}: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$$\bar{f} \text{ не є в } \Gamma: x_0, \text{ єдином } \lim_{x \rightarrow x_0} \bar{f}(x) = \bar{f}(x_0)$$

$$\text{Def} \quad \bar{f} \text{ не є на } (a; b), \text{ єдином } \forall x \in (a; b) \quad \bar{f} \text{ не є в } x$$

$$\text{Def} \quad \bar{f}: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$$\bar{f}'(x) = \lim_{x \rightarrow x_0} \frac{\bar{f}(x) - \bar{f}(x_0)}{x - x_0}$$

$$\text{Ymb: } \exists \bar{f}'(x_0) \Leftrightarrow \exists f'_k(x_0) \quad \forall k = \{1, \dots, n\}$$

$$\text{Def.} \quad \bar{f}: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$$\varphi: \dot{U}_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$$\lim_{x \rightarrow x_0} \frac{\bar{f}(x)}{\varphi(x)} = \bar{o}$$

$$\left(\bar{f} \text{ д.н. омн. } \varphi \text{ при } x \rightarrow x_0 \right)$$

$$\bar{f}(x) = \bar{o}(\varphi(x)) \quad x \rightarrow x_0$$

$$\text{Ymb: } \bar{f}(x) = \bar{o}(\psi(x)) \quad x \rightarrow x_0$$

$$f_k(x) = o(\psi(x)) \quad x \rightarrow x_0 \quad \forall k \in \{1, \dots, n\}$$

$$\text{Def: } \bar{f}: U_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$\bar{f}$ глуп б т. x_0 , ели $\exists \bar{a}$:

$$\bar{f}(x) - \bar{f}(x_0) = \bar{a}(x - x_0) + \bar{o}(x - x_0)$$

$\bar{f}(x_0) = f'(x_0) \cdot x$ мого и глуп елима могоб ел. во могоб

$$\text{Ymb: } \bar{f}: U_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$\bar{f}$ глуп б т. $x_0 \Rightarrow \bar{f}$ мого б т. x_0

$$\text{Ymb: } \bar{f}; \bar{g}: U_{\delta_0}(x_0) \rightarrow \mathbb{R}^n$$

$\bar{f}; \bar{g}$ глуп б т. x_0

$$\psi: U_{\delta_0}(x_0) \rightarrow \mathbb{R}$$

ψ глуп б т. x_0

$$(\bar{f} + \bar{g})' = \bar{f}' + \bar{g}'$$

$$(\bar{f}; \bar{g})' = (\bar{f}' \bar{g}) + (\bar{f}; \bar{g}')$$

ели $n=3$

$$[\bar{f}; \bar{g}]' = [\bar{f}', \bar{g}] + [\bar{f}; \bar{g}']$$

$$(\psi \bar{f})' = \psi' \bar{f} + \psi \bar{f}'$$

$$\lim_{x \rightarrow x_0} \frac{[\bar{f}(x); \bar{g}(x)] - [\bar{f}(x_0); \bar{g}(x_0)]}{x - x_0} =$$

$$= \lim_{x \rightarrow x_0} ([\bar{f}(x); \bar{g}(x)] - [\bar{f}(x_0); \bar{g}(x_0)]) + ([\bar{f}(x_0); \bar{g}(x)] - [\bar{f}(x_0); \bar{g}(x_0)]) =$$

$$= \lim_{x \rightarrow x_0} \left(\frac{[\bar{f}(x) - \bar{f}(x_0), \bar{g}(x)]}{x - x_0} + \frac{[\bar{f}(x_0), \bar{g}(x) - \bar{g}(x_0)]}{x - x_0} \right) =$$

$$= \lim_{x \rightarrow x_0} \left(\left[\frac{\bar{f}(x) - \bar{f}(x_0)}{x - x_0}, \bar{g}(x) \right] + \left[\bar{f}(x_0), \frac{\bar{g}(x) - \bar{g}(x_0)}{x - x_0} \right] \right) =$$

$$= [\bar{f}'(x_0); \bar{g}(x_0)] + [\bar{f}(x_0); \bar{g}'(x_0)] \blacktriangle$$

$$\text{Ymb: } \psi: U_{\delta_0}(x_0) \rightarrow \mathbb{R}$$

$$\psi(x_0) = y_0$$

$$\bar{f}: U_{\delta_0}(y_0) \rightarrow \mathbb{R}^n$$

ψ глуп б т. x_0

$\bar{f}$ глуп б т. y_0

$$\bar{h}(x) = \bar{f}(\psi(x))$$

глуп б т. x_0

$$\bar{h}'(x_0) = \bar{f}'(y_0) \cdot \psi'(x_0)$$

$$\text{Ymb: } \bar{f}: [a; b] \rightarrow \mathbb{R}^n$$

$\bar{f}$ мого на $[a; b]$

$\bar{f}$ глуп на $[a; b]$

$$\Rightarrow \exists \xi \in (a; b) : |\bar{f}(b) - \bar{f}(a)| \leq |\bar{f}'(\xi)| |b - a|$$



► $\psi(x) = (\bar{f}(x), \bar{f}(b) - \bar{f}(a))$

$\psi: [a; b] \rightarrow \mathbb{R}$

ψ непрерывна на $[a; b]$

ψ убывает на $(a; b)$

уже концы то?!